

Propositional Logic: Strength and Weakness

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$$B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{1,3} \vee P_{2,2})$$

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- ♠ Propositional logic assumes the world contains *facts only*.

Combining Formal and Natural Languages

First-order logic

- ◆ built around objects and relations
 - **Objects:** people, houses, cars, trees, colors, days, ...
 - **Relations:**
 - ◆ unary properties such as `big`, `windy`, ...
 - ◆ *n*-ary properties such as `bigger than`, `parent of`, `on`, `owns`, ...
 - **Functions:** `square of`, `best friend`, `age`, ...
- ◆ capable of expressing facts about some or all objects

Formal Languages

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Alphabet of First-Order Logic

◆ Logical symbols

- connectives: $\wedge, \vee, \Rightarrow, \Leftrightarrow, \neg$
- parenthesis: $(,)$ and punctuation $,$
- equality: $=$
- quantifiers: $\forall$ (universal quantification), $\exists$ (existential quantification)
- variables: $x, y, z, \dots; x_1, x_2, \dots$

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- constants: Socrates, Turing, 1, earth, ...

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 - true, false*
 - Father*(x, y) // x is father of y
 - Female*(x) // x is female

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- predicate symbols:
 - true*, *false*
 - Father*(x, y) // x is father of y
 - Female*(x) // x is female
- function symbols:
 - gcd*(x, y) // greatest common divisor of x and y
 - FatherOf*(x) // father of x

Terms and Atomic Sentences

◆ Terms

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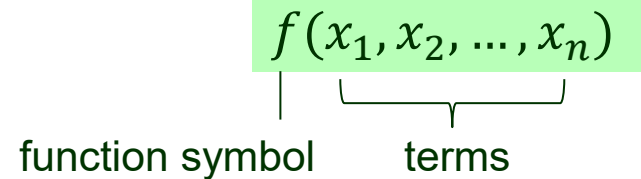
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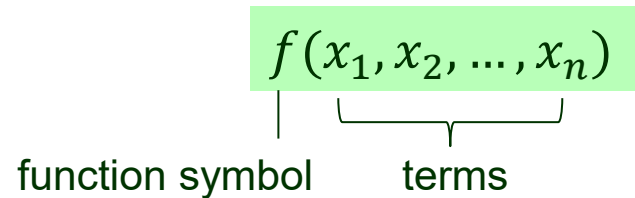
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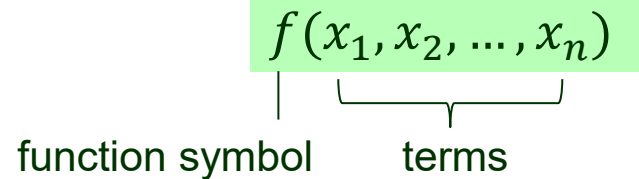
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- predicates: *true*, *false*

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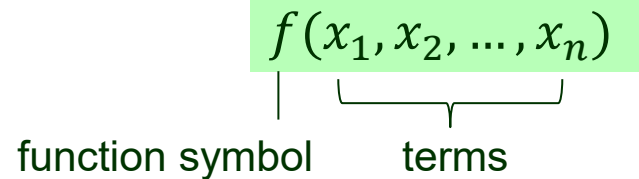
◆ Atomic sentences

- predicates: *true*, *false*
Mother(Aphrodite, Harmonia)
Male(John)

Terms and Atomic Sentences

◆ Terms

- constants: Socrates, Turing, 1, earth, ...
- variables: $x, y, z, \dots$; $x_1, x_2, \dots$
- functions: $\text{gcd}(x, y)$, $\text{FatherOf}(x)$, ...



◆ Atomic sentences

- predicates: *true*, *false*
 $\text{Mother}(\text{Aphrodite}, \text{Harmonia})$
 $\text{Male}(\text{John})$
- term equalities

$$\text{FatherOf}(\text{Apollo}) = \text{Zeus}$$

Complex Sentences

- made of atomic sentences using logical connectives

Father(x, y) $\Rightarrow$ *Male*(x)

Female(x) $\vee \neg$ *Mother*(x, y)

Likes(Mary, John) $\Leftrightarrow$ *Likes*(John, Mary)

(*Parent*(x, y) $\wedge$ *Parent*(y, z)) $\Rightarrow$ *GrandParent*(x, z)

Complex Sentences

- made of atomic sentences using logical connectives

$Father(x, y) \Rightarrow Male(x)$

$Female(x) \vee \neg Mother(x, y)$

$Likes(Mary, John) \Leftrightarrow Likes(John, Mary)$

$(Parent(x, y) \wedge Parent(y, z)) \Rightarrow GrandParent(x, z)$

- universal quantification

$\forall x \ Circle(x) \Rightarrow Ellipse(x)$ // Every circle is an ellipse.

$\neg \forall x \ Likes(x, sushi)$ // Not everyone likes sushi.

$\forall x \ Integer(x) \Rightarrow (Even(x) \vee Odd(x))$ // Every integer is either even or odd.

Complex Sentences

- made of atomic sentences using logical connectives

$Father(x, y) \Rightarrow Male(x)$

$Female(x) \vee \neg Mother(x, y)$

$Likes(Mary, John) \Leftrightarrow Likes(John, Mary)$

$(Parent(x, y) \wedge Parent(y, z)) \Rightarrow GrandParent(x, z)$

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$\forall x \text{ Integer}(x) \Rightarrow (\text{Even}(x) \vee \text{Odd}(x))$ // Every integer is either even or odd.

- existential quantification

$\exists x \text{ Star}(x) \wedge \neg (x = \text{Sun})$ // There are stars other than the sun.

$\exists x \text{ Whale}(x) \wedge (\text{Age}(x) = 200)$ // Some whales live to 200 years.

Syntax of First-Order Logic

Sentence $\rightarrow$ *AtomicSentence* | *ComplexSentence*

AtomicSentence $\rightarrow$ *Predicate* | *Predicate*(*Term*,...) | *Term* = *Term*

ComplexSentence $\rightarrow$ (*Sentence*)

| $\neg$ *Sentence*

| *Sentence* $\wedge$ *Sentence*

| *Sentence* $\vee$ *Sentence*

| *Sentence* $\Rightarrow$ *Sentence*

| *Sentence* $\Leftrightarrow$ *Sentence*

| *Quantifier* *Variable*,... *Sentence*

Term $\rightarrow$ *Function*(*Term*,...)

| *Constant*

| *Variable*

Quantifier $\rightarrow$ $\forall$ | $\exists$

Constant $\rightarrow$ *A* | *X*₁ | *John* | ...

Variable $\rightarrow$ *a* | *x* | *s* | ...

Predicate $\rightarrow$ *True* | *False* | *After* | *Loves* | *Raining* | ...

Function $\rightarrow$ *Mother* | *LeftLeg* | ...

OPERATOR PRECEDENCE : $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Translating English Sentences into FOL

1. Everyone makes mistakes.

$$\forall x \exists y \text{ Human}(x) \Rightarrow (\text{Does}(x, y) \wedge \text{Mistake}(y))$$

2. One person is an ancestor of another if he is a parent of the other or he is an ancestor of an ancestor of the other.

$$\forall x \forall y \text{ Parent}(x, y) \vee \exists z (\text{Ancestor}(x, z) \wedge \text{Ancestor}(z, y)) \Rightarrow \text{Ancestor}(x, y)$$